LeoWay — Project Proposal

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Summary

LeoWay is a commute assistant for HTL Leonding that connects each student's school timetable with live public transport data. It tells students when to leave based on their actual first lesson, including cancellations and substitutions, and warns them early when their connection is delayed or students report a disruption. When a student is marked late because of a delay, LeoWay drafts an appeal email to the form teacher with a verifiable delay record, replacing today's photos and Scotty screenshots. The delay data it records over time trains a prediction model and gives the school evidence about which connections make students late.

Initial situation

Many students at HTL Leonding arrive late, and public transport delays are a frequent cause. Three things make it worse:

- **Transit apps don't know the school timetable.** Scotty or the Linz AG app show when a bus leaves, but not when the student's first lesson starts, or that it was cancelled or moved.
- **Delays are noticed too late.** Students find out at the stop, when it's too late to take an earlier connection.
- **Excusing lateness is manual.** A late student photographs the screen in the bus, tram or train, or takes a Scotty screenshot, and sends it to the form teacher. The teacher can't check whether it's genuine or from that morning, and may decline the request if the student took a connection that is often late.

Goals

LeoWay succeeds if students leave on time more often and teachers can trust delay appeals. The project has no fixed yearly plan; these goals feed the product backlog.

Goal	Target
Timetable-aware alerts	Every alert accounts for the student's actual first lesson, including cancellations and substitutions
Delay appeals	A late student drafts an appeal email to the form teacher in under a minute; the teacher verifies it from the link inside
Data history	The delay recorder runs continuously from the first sprint, including holidays
Crowd reports	A disruption that several students confirm triggers an alert before official feeds show it
Prediction	The model beats two baselines, "same delay as yesterday" and "average delay of this trip"
School statistics	One report on connections that regularly arrive after lessons start, handed to the student council

Features

Features are not split by year; all of them go into the product backlog, and the team works through them in the order under Backlog. Most of the work is backend: data adapters, the recorder, matching, prediction and integrations.

For students

- **Leave-by alerts:** reads the student's timetable from WebUntis, combines it with their saved route and says when to leave, e.g. "first lesson cancelled, take the 7:45 bus".
- **Live delays:** watches the student's connection in the morning, warns early and suggests an alternative when a delay makes them late.
- **Alarm:** on Android, a notification at the exact leave-by time that works like an alarm, e.g. moved later when the first lesson is cancelled.
- **Way home:** the next connection after the student's actual last lesson, including when lessons end early.
- **Custom alert rules in auspex-lang**, e.g. "notify me if the delay is over 5 minutes and my first lesson is a test".

Delay appeals

- **Absence matching:** LeoWay reads the student's lateness entries in WebUntis (short absences at the start of the day) and matches them with its recorded delays, e.g. "You were marked 13 min late; your bus 19 was 11 min late." It only reads WebUntis and ignores absence reasons and notes.
- **Appeal drafts:** one tap drafts an email to the form teacher with line, stop, scheduled and actual time, minutes late and a verification link. The student checks it and sends it from their own account.
- **Teacher portal (optional):** teachers who enroll handle appeals in the portal instead of by email: verify, approve or decline, with the line's reliability shown next to each. Teachers who don't enroll simply keep getting emails.

Data and prediction

- **Delay recorder:** stores every departure and delay from the first sprint on, as the basis for prediction and statistics.
- **Crowd reports:** students report disruptions in a few taps; once several confirm, LeoWay flags an outage and alerts everyone on that line, often before official feeds show it.
- **Prediction:** a risk score for tomorrow's connection, a personal buffer suggestion ("leave 6 minutes earlier on Mondays") and an extra warning when the first lesson is a Schularbeit.
- **School statistics:** connections that arrive after lessons start, and lateness hotspots, for the student council to take to Linz AG or OÖVV.

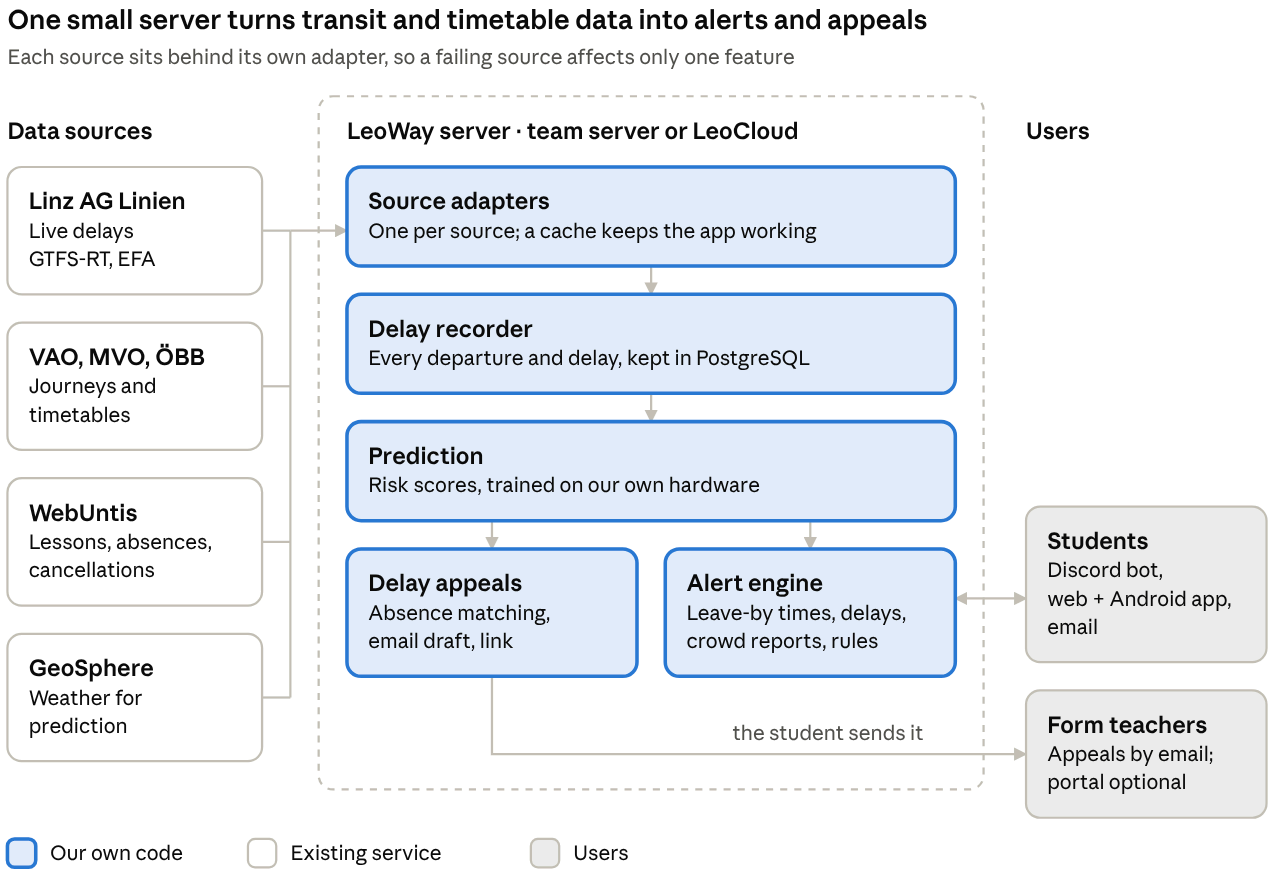
Integrations

- **Channels:** Discord bot first, then push notifications and email; further chat platforms and school portals plug into the same notification interface.
- **Later:** other schools in Linz, an anonymized open delay dataset, and indoor navigation from the stop to the classroom.

Out of scope

- **Writing to WebUntis:** LeoWay never submits absences or excuses; it only reads.
- **Location tracking:** no GPS, no home address.
- **A native iPhone app:** iPhone users get the website.
- **Proving a student was on the vehicle:** LeoWay proves the delay; the teacher still decides about the appeal.

How it works



LeoWay architecture · 4 sources, 5 components, 2 user groups

Every source goes through its own adapter into the delay recorder; prediction, alerts and delay appeals all build on that recorded history. The stack is Quarkus for the backend, Angular for the student app and teacher portal, PostgreSQL for data, and Python for training the model on our own hardware.

App or website: the Angular app runs as a website and is wrapped with Capacitor into an Android app. A website can't schedule exact-time alarms and is limited on iPhones; with Capacitor one codebase serves both, and the Android app can fire the leave-by alarm.

Data sources

The last leg to school is covered by open live data today; regional buses and trains need the VAO license for live data and use timetables until then. HTL Leonding is served by Linz AG bus 19 at Limesstraße and by trams 3 and 4 and buses 11, 19 and 191 at Meixnerkreuzung, plus regional OÖVV buses such as 611 and 618.

Source	Data	Access	Status
Linz AG Linien	Live departures and delays (GTFS-RT, EFA departure endpoint)	Free	Available
Mobilitätsverbünde Österreich (MVO)	Timetables for regional buses incl. OÖVV (GTFS)	Free registration	Available
ÖBB	Train timetables (GTFS, CC BY 4.0)	Free	Available
VAO Start	Journeys and departures across Austria	Contract; free tier 100 queries per day per service; discounts for education	Education license to be requested
WebUntis	Student's timetable, cancellations, substitutions, lateness entries	Student's personal access key (Profil → Freigaben), via the `webuntis` npm package; no password needed	Tested with a real student account: timetable, cancellations, substitutions, exams and absences all load
GeoSphere Austria	Weather, as input for prediction	Free	Available

Each source sits behind its own adapter, so a changed or failed source affects one feature, not the whole app.

Verified with WebUntis

On 2026-10-08 we tested HTL Leonding's WebUntis (`htl-leonding.webuntis.com`, school `htl-leonding`, ID 7049600) with a real student account and the `webuntis` npm package 2.2.1. Everything LeoWay needs loads; anonymous access is disabled, so each student must connect their own account.

Capability	How	Result
Login without password	Username + access key; the package generates a one-time code (`WebUntisSecretAuth` , needs `otplib` 12)	Works, about 0.6 s; repeatable every morning
Weekly timetable	JSON-RPC timetable for the student	38 lessons, under 150 ms
Cancellations	Lesson field `code` : `cancelled` or `irregular`	E.g. 08.10.: 08:00 cancelled, first real lesson 08:55
Substitutions	Teacher entry with `orgname` = the original teacher	Visible per lesson
Lesson times	Timegrid	Period 0 starts 07:05, period 1 at 08:00
Absences and lateness	`getAbsentLesson`	Date, start and end time, reason, note, excuse status; lateness appears as a short absence, e.g. 08:00–08:13
Exams, holidays, school year	Separate calls	All load
Newer REST API	JWT from the session	Works

Not tested yet: looking up the form teacher's email, rate limits. The test scripts are in `webuntis-spike/` (`probe.mjs` , `absences.mjs`).

Privacy

LeoWay is voluntary, tracks no location and stores only what the features need. In Austria, students aged 14 and over can consent to an app like this themselves.

Data	Stored	Kept until
Saved route (home stop, not home address)	Yes	Account deleted

Data	Stored	Kept until
Timetable from WebUntis	Cached	Replaced on next sync; deleted with the account
WebUntis access key	Yes, encrypted	Account deleted, or key revoked by the student in WebUntis
Delay appeals (line, time, delay)	Yes	End of school year
Lateness entries from WebUntis	Only date, time and the matching appeal; absence reasons and notes are never stored	End of school year
Discord ID, push token, email	Yes, for alerts	Account deleted
WebUntis password	No	–
GPS location, home address	No	–

The delay history of buses and trams is not personal data. Teachers only see a delay appeal when the student sends it, and class statistics in the teacher portal are aggregated.

Team and roles

Work is assigned per sprint, not by fixed area, so everyone works on backend, apps and data over the course of the project.

Member	Role
Laurenz	Scrum master and developer; runs the GitHub repository: writes issues, assigns work, reviews and merges pull requests
Ayan	Developer
David I.	Developer

Member	Role
Behrang	Developer
Product owner	Open: the supervising teacher is the customer but doesn't take this role

Way of working

We work in Scrum with two-week sprints across both school years.

- **Roles:** the supervising teacher is the customer; Laurenz is Scrum master; the product owner is still open.
- **Stakeholders in sprint reviews:** students from our own classes, and at least one form teacher once delay appeals exist.
- **Backlog:** user stories grouped into the epics under Backlog, as GitHub issues in GitHub Projects; Laurenz assigns them each sprint. The team communicates on Discord.
- **Definition of done:** code reviewed, tests passing in CI, deployed to the test server and documented.
- **Holidays:** no sprints in school holidays; the delay recorder keeps running.

Backlog

There is no fixed release plan. The epics below form the product backlog in priority order; each sprint takes the next stories from the top, and the order is reviewed in every sprint review.

1. **Delay recorder and transit adapters**, running from the first sprint so the data history starts early
2. **WebUntis access-key login and leave-by alerts**
3. **Student app** (Angular website, Android app via Capacitor) and the **Discord bot**
4. **Live delays** with alternative connections
5. **Delay appeals:** absence matching and the appeal email draft
6. **Crowd reports** and outage detection
7. **Prediction:** baselines first, then the model and risk scores
8. **Optional teacher portal** for handling appeals
9. **More integrations:** further chat platforms and school portals

10. **School statistics, way home and auspex-lang rules**
11. **Later:** other schools, open delay dataset, indoor navigation

Risks

No single failure stops the project; each risk has a fallback that keeps the core working.

Risk	Effect	Fallback
Untis or the school disables access keys or the WebUntis JSON-RPC API	No automatic timetable import	iCal link or manual entry behind the same adapter; ask Untis about a platform app
VAO license is declined or too limited	No live data for regional buses and trains	Live data for Linz AG lines, timetables for the rest; cached queries
A feed is down or changes format	One source missing	Cached data, a status notice in the app, and an alert to the team
The prediction model doesn't beat the baselines	Weaker risk scores	The baselines run in the product anyway; the comparison is documented as a result
Few students try LeoWay	Little feedback	Start with our own classes; the Discord bot needs no install
Teachers don't accept delay appeals	Appeal feature unused	Agree the format with form teachers before building it; the verification link is the selling point
Android restricts exact-time alarms for the app	Leave-by alarm fires late	Test it in an early sprint; fall back to a high-priority push notification

Next steps

Before the first sprint we settle the open points that could change the scope.

- [ ] Apply for a VAO Start education license
- [ ] Short survey in two or three classes: how do you get to school, how often are you late, and why?
- [ ] Ask school IT whether students may use their WebUntis access key for LeoWay, and whether the school uses digital absence reports in WebUntis
- [ ] Ask one or two form teachers what a delay appeal must show for them to accept it
- [ ] Find a product owner
- [ ] Test whether a Capacitor app can fire an exact-time alarm on Android

For the system spec

The proposal fixes what LeoWay does and why; the system spec decides how. It should use these terms and answer the questions below.

Term	Meaning
Leave-by time	When a student must leave home to reach their first real lesson, from timetable, route and live delays
Delay record	One recorded departure: line, stop, scheduled and actual time, as stored by the delay recorder
Lateness entry	A short WebUntis absence at the start of the day, meaning the student arrived late
Delay appeal	An email the student sends to the form teacher, matching a lateness entry with a delay record
Verification link	A link in the appeal that shows the teacher LeoWay's own delay record
Crowd report	A student's report of a disruption; several confirmations make an outage
Access key	The student's WebUntis key (Profil → Freigaben), used instead of the password

Open questions

1. How does LeoWay find the form teacher's email: from WebUntis, from the school directory, or entered by the student?
2. What does the verification link show, how long is it valid, and how does it prevent forged records?
3. Matching: which time window and which line or stop link a lateness entry to a delay record?
4. Crowd reports: how many confirmations make an outage, and how are false reports limited?
5. How often are timetables and absences synced, and how does LeoWay stay within WebUntis load limits?
6. How are access keys encrypted, and where does the encryption key live?
7. How do teachers log in to the optional portal?
8. What does the shared notification interface look like, so Discord, push, email and later platforms plug in the same way?
9. Hosting: the team's own server or LeoCloud?

Sources

- Offene Öffis: open transport data in Austria
- MVO data platform
- VAO Start
- linz-linien-austria, uses the Linz AG live endpoint
- webuntis npm package
- WebUntis release notes (2FA and the old API)
- Untis platform developer documentation
- Stadt Leonding: public transport
- Linzwiki: HTL Leonding
- GeoSphere Austria Data Hub